

FRAME FOR PERFORMANCE

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-451 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Frame with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.

ADVANCED RESEARCH MISSION

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-452 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Design with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.

INSTRUMENT THE DESIGN

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-453 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Instrument with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.

CALIBRATE FOR PERFORMANCE

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-454 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Calibrate with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.

ADVANCED RESEARCH MISSION

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-455 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Model with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.

VALIDATE THE DESIGN

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-456 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Validate with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.

EVALUATE FOR PERFORMANCE

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-457 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Evaluate with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.

ADVANCED RESEARCH MISSION

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-458 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Optimize with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.

STRESS-TEST THE DESIGN

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-459 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Stress-test with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.

RECOMMEND FOR PERFORMANCE

SSS 1-3 | Advanced Soilless Garden

Frame and execute a research or engineering mission on recirculating soilless control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT pH/EC meters, flow data, system map

EVIDENCE pH/EC dynamics, residence time or diagnostic evidence

GM-460 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Recommend with pH/EC meters, flow data, system map. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for pH/EC dynamics, residence time or diagnostic evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Teacher supervises nutrient-solution work. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Advanced Soilless Garden, including a measurable acceptance criterion.